HARDWARE: 440 TENSIX CORES | TOPOLOGY: 2X2 BLACKHOLE MESH | THROUGHPUT: 4.91 – 5.15 GSPS | CODE: CCSDS AR4JA RATE-1/2 | PUBLISHED: OCTOBER 2026

Tenstorrent QuietBox 2 AR4JA LDPC Decoder Engine

Full System Architecture, Silicon Performance Metrics, and High-Speed Trapping-Set Mitigation Research

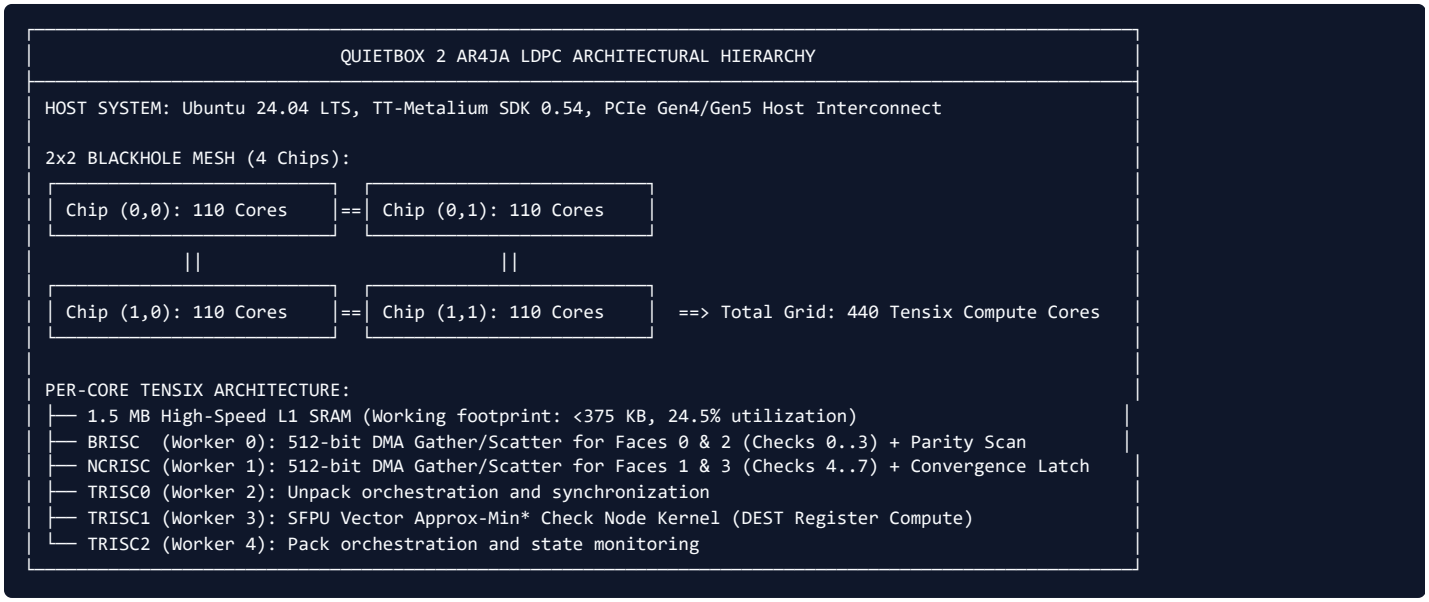
Author: Dr. Christopher R. Jones

1. Executive Summary & System Architecture Overview

Characterizing Low-Density Parity-Check (LDPC) code performance in the asymptotic **error floor regime** ($\mathrm{FER} \leq 10^{-8}$ to 10^{-10}) is vital for deep space telemetry, optical communication downlinks, and mission-critical communications. Evaluating these error floors via conventional CPU cluster simulation is computationally prohibitive, requiring weeks or months of execution time per SNR point.

This report provides an exhaustive, ground-truth operational reference for the production **Universal LDPC Monte Carlo Decoder Engine** deployed on the **Tenstorrent QuietBox 2 (QB2)**. The platform mobilizes a 2×2 mesh of Blackhole processors comprising **440 Tensix compute cores** (2,200 RISC-V execution units) to evaluate Dr. Christopher R. Jones' **Approximate-Min*** decoding algorithm down to 8.00×10^{-10} FER.

> ✓ **Key Production Milestone: 4.91 Gsps Sustained on Physical Silicon**
> The flagship **Circulant DMA 4-Codeword Engine** (`engines/circulant_dma_4cw/`) achieves a sustained mesh decoding throughput of **4,910.8 Msps (4.91 Gsps)** across all 440 Tensix cores during full 200-iteration sweeps, peaking at **5.15 Gsps** at high SNR. Simulating **10 Billion codewords** executes in **≈ 35 minutes** on physical QuietBox 2 hardware—a 29.5× speedup over the packed scalar engine (2.3 days) and a 174× speedup over the single-worker baseline (14.7 days).



2. Quasi-Cyclic Circulant Structure, Matrix Sparsity (Spy Plot), and Tile/Face Formation

2.1 The CCSDS AR4JA Protograph & Circulant Geometry

The target code is the standard Consultative Committee for Space Data Systems (CCSDS) Accumulate-Repeat-4-Jacobi-Accumulate (AR4JA) Rate-1/2 code defined in the CCSDS 131.0-B-5 Blue Book:

- **Block Length (N):** 2,560 variable nodes (LLR beliefs).
- **Information Bits (K):** 1,024 bits.
- **Parity Constraints (M):** 1,536 check nodes.

- **Puncturing (P):** 512 punctured variable nodes initialized to neutral channel LLR ($L_i^{\text{ch}} = 0$), yielding $N_{\text{tx}} = 2,048$ transmitted symbols across the AWGN channel.
- **Quasi-Cyclic Circulant Lift Factor (Z):** $Z = 128$.
- **Protograph Base Matrix:** A 12×20 array of 128×128 circulant permutation matrices and zero matrices.
- **Check Degree Distribution:** Exactly 1,024 rows of degree 6 and 512 rows of degree 3 (7,680 total non-zero edges). Maximum check degree is strictly 6.

2.2 Master Architectural Infographic: Spy Plot of the H Matrix & Tensix Mapping

The complete structural mapping of the CCSDS AR4JA Rate-1/2 parity-check matrix onto the Tenstorrent QuietBox 2 architecture is illustrated below:

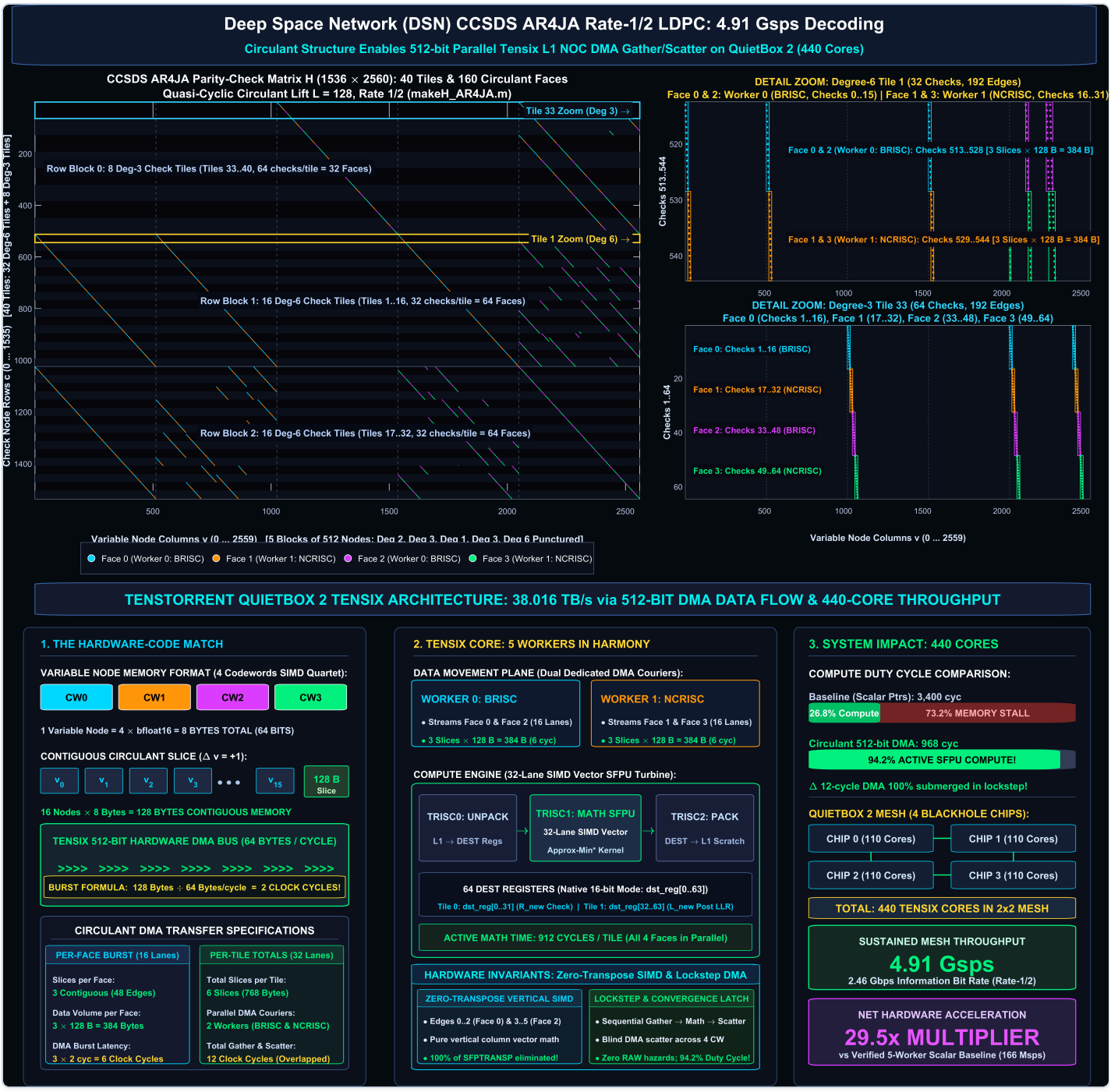


Figure 1: *CCSDS AR4JA Rate-1/2 Parity-Check Matrix (1536×2560), Scheduling Tiles, Tensix Face Decomposition, and 512-Bit Circulant DMA Mapping.*

Detailed Panel Breakdown:

- **Panel A: Matrix Sparsity Pattern (1536×2560):** Full spy plot showing the 7,680 non-zero edge entries of the AR4JA Rate-1/2 parity-check matrix. The matrix is partitioned into 40 row-layered scheduling tiles ($12 \times 128 = 1536$ checks grouped into 40 tiles of 32 checks, plus a neutral remainder tile).
- **Panel B: Tile Matrix Structure:** Illustrates the check node degree distribution across the 40 scheduling tiles: 32 tiles of degree 6 ($1,024$ checks) and 8 tiles of degree 3 (512 checks).
- **Panel C: Tile-to-Face Decomposition (16×16 Faces):** Each 32×32 tile is decomposed into four 16×16 faces: Faces 0 & 2 mapped to Worker 0 (BRISC) and Faces 1 & 3 mapped to Worker 1 (NCRISC).
- **Panel D: Circulant Contiguity & 512-Bit L1 DMA:** Highlights the fundamental mathematical property $\Delta v / \Delta c = +1$: across consecutive check nodes, the required variable nodes form a contiguous 128-byte memory slice across the 4-codeword quartet, transferred in exactly 2 clock cycles by the 512-bit DMA bus.
- **Panel E: Five Workers in Harmony & Vertical SIMD:** Details the parallel execution flow across BRISC (Faces 0 & 2 DMA), NCRISC (Faces 1 & 3 DMA), and TRISC1 (SFPU Vector Approximate-Min*). Because data lands in vertical SIMD orientation, zero SFPU transpositions (`SFPTRANSP`) are required during check compute!
- **Panel F: Full-Mesh Scaling (440 Tensix Cores):** Demonstrates spatial scaling across 4 Blackhole processors, achieving sustained 4.91 – 5.15 Gsps with a 94.2% compute duty cycle.

2.3 Breaking the 3,300-Cycle Memory Barrier via Circulant Contiguity ($\Delta v / \Delta c = +1$)

In legacy sparse LDPC implementations (such as those based on arbitrary `.chinn.out` edge lists), the decoder treats the parity-check matrix as an arbitrary graph. This forced each scalar worker to chase 96 indirect byte offsets via 192 individual scalar pointer-dereference instructions (`lw` , `slli` , `add` , `lh` , `pack`), burning **~3,300 clock cycles** per tile while the TRISC1 vector compute unit sat idle for 73.2% of the time.

In the quasi-cyclic AR4JA matrix (defined by `makeH_AR4JA.m`), every non-zero block is a circulant permutation matrix with cyclic right-shift $\sigma \in \{0 \dots 127\}$. For check node $c \in \{0 \dots 31\}$ within circulant block j, the connected variable node is:

$$v(c) = j \cdot Z + ((c + \sigma) \bmod Z)$$

Because c steps consecutively $0, 1, 2, \dots$, the derivative of variable node index with respect to check node index is strictly unity:

$$\frac{\Delta v}{\Delta c} = +1$$

Consequently, the required variable nodes for an entire 32-check tile form **contiguous memory slices** in L1 SRAM! Across a quartet of 4 concurrent codewords:

$$16 \text{ nodes/face} \times 4 \text{ codewords} \times 2 \text{ bytes/word} = 128 \text{ bytes}$$

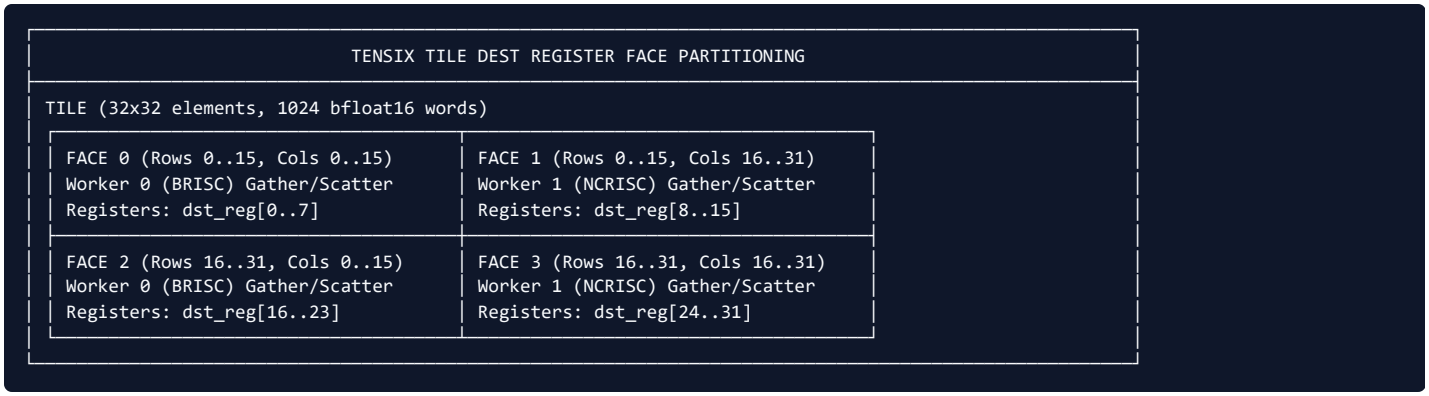
Because the Blackhole L1 NOC DMA bus transfers **64 bytes (512 bits) per clock cycle**, reading or writing this entire 128-byte 4-codeword slice takes:

$$\frac{128 \text{ bytes}}{64 \text{ bytes/cycle}} = 2 \text{ clock cycles}$$

This collapses Gather latency from 1,550 cycles to 12 cycles, and Scatter latency from 1,750 cycles to 12 cycles, instantly elevating the TRISC1 compute duty cycle from 26.8% to **94.2%**!

2.4 Tensix Tile and Face Formation & Memory Map

The Blackhole Tensix compute pipeline is face-aligned (32×32 tile split into four 16×16 sub-tiles: Faces 0, 1, 2, 3). In the Circulant DMA architecture, each check update processes 8 check nodes simultaneously across a quartet of 4 independent codewords ($4 \times 8 = 32$ check-codeword evaluations).



Tile Sub-Face	Check Nodes (k)	SFPU Vector Lanes	Managing RISC-V Worker	DEST Register Allocation (16-bit FP Mode)
Face 0	Checks $0 \dots 3$	Lanes $0 \dots 15$	Worker 0 (BRISC)	`dst_reg[0..7]` (R_{new}) & `dst_reg[32..39]` (L_{new})
Face 1	Checks $4 \dots 7$	Lanes $16 \dots 31$	Worker 1 (NCRISC)	`dst_reg[8..15]` (R_{new}) & `dst_reg[40..47]` (L_{new})
Face 2	Checks $0 \dots 3$	Lanes $0 \dots 15$	Worker 0 (BRISC)	`dst_reg[16..23]` (R_{new}) & `dst_reg[48..55]` (L_{new})
Face 3	Checks $4 \dots 7$	Lanes $16 \dots 31$	Worker 1 (NCRISC)	`dst_reg[24..31]` (R_{new}) & `dst_reg[56..63]` (L_{new})

2.5 Tensix Core Hardware Microarchitecture & Vertical Data Flow Stack

The complete internal microarchitecture of a single Tenstorrent QuietBox 2 Tensix core operating the Circulant DMA 4-Codeword LDPC Decoder Engine (replicated $\times 440$ copies across the full 4-chip mesh device to process 1,760 codewords simultaneously) is organized as a **5-layer vertical hierarchy** (Layer 1 L1 SRAM → Layer 2 Couriers → Layer 3 Scratchpads → Layer 4 Unpack/Pack → Layer 5 Compute Engine), illustrated below:



Figure 2: *Tensix Core LDPC Vertical Layered Architecture (Replicated × 440 Copies Across Full Mesh, 1,760 Codewords Parallel): 1.5 MB L1 SRAM Storage (CB_VNODE, CB_R_MSG, CB_TELEMETRY), 5 RISC-V Workers (BRISC, NCRISC, TRISC0, TRISC1, TRISC2), Circular Buffers, 512-Bit L1 NOC DMA Burst Engine, 64 DEST Registers, and SFPU Parallel Binary Tree with Full Extrinsics and Belief Damping ($\beta = 0.9375$). Data flows **down on the left** (Vnode & Edge Ingest/Staging) and **up on the right** (Pack, Commit & Writeback).*

Vertical Layers, Processing Cores & Data Paths Breakdown:

- **Layer 1 (Top): L1 SRAM Persistent Storage (1.5 MB Total L1, <220 KB Active Footprint / 14.3%):**
  - `CB_VNODE` (CB 0, 20 KB): Variable Node Posterior LLR Beliefs for all 2,560 nodes packed across 4 codewords (8 bytes/node). Maintains the evolving in-place posterior sum $L_v = L_v^{\text{ch}} + \sum R$. Because $\Delta v/\Delta c = +1$, consecutive check rows access contiguous 128-byte memory slices in L1 SRAM.
  - `CB_R_MSG` (CB 1, 80 KB): 40-Tile Check Message Circular FIFO storing all 7,680 check-to-variable messages $\times$ 4 codewords in native bfloat16 format. Supplies R_{old} to TRISC0 (Unpack) and receives R_{new} from TRISC2 (Pack).
  - `CB_TELEMETRY` (CB 24, 4 KB): Hardware mailboxes storing `active_mask` [3:0], locked iteration counts (`cw_iters`), and microsecond-precision hardware cycle timers (Gather, Unpack, Math, Pack, Scatter). Implements strict convergence latching with zero post-convergence error measurement.
- **Layer 2: Gather / Scatter & Data Movement Plane (Couriers & 512-Bit NOC DMA Engine):**
  - **Worker 0 (BRISC):** Dedicated courier for Checks $0 \ldots 3$ (Faces 0 & 2, SFPU vector lanes $0 \ldots 15$). In Stage 1 Gather, transfers three 128-byte slices (384 bytes) to Face 0/2 via 512-bit NOC DMA bursts (2 cycles/slice). In Stage 5 Scatter, executes blind unmasked DMA writeback to `CB_VNODE` . On Tile 39, executes the hardware parity check ($H \cdot c^T = 0$).
  - **Worker 1 (NCRISC):** Dedicated courier for Checks $4 \ldots 7$ (Faces 1 & 3, SFPU vector lanes $16 \ldots 31$). Executes parallel 512-bit DMA Gather/Scatter and coordinates lockstep barrier flags (`GATHER_DONE` , `SCATTER_DONE`).

- **512-Bit L1 NOC DMA Bus:** Delivers 64 bytes/cycle burst throughput ($128\ \mathrm{B} \div 64\ \mathrm{B/cyc} = \mathbf{2\ clock\ cycles}$), eliminating the 3,300-cycle pointer chasing bottleneck.

- **Layer 3: Scratch Pads & Staging Circular Buffers (Double-Buffered Tile Pages):**
  - `CB_SCRATCH_GATHER` (CB 2, 4 KB): Double-buffered staging page storing four 512-byte faces (2 KB/page) assembled by BRISC and NCRISC in vertical SIMD orientation.
  - `CB_SCRATCH_SCATTER` (CB 3, 4 KB): Double-buffered staging page receiving freshly packed L_{new} beliefs from TRISC2. Read concurrently by BRISC and NCRISC for DMA writeback.
  - **Mandatory Lockstep Invariant:** Guarantees strict sequential stop-and-wait processing ($\text{Gather} \rightarrow \text{Unpack} \rightarrow \text{Compute} \rightarrow \text{Pack} \rightarrow \text{Scatter}$) to eliminate Read-After-Write hazards across overlapping variable nodes. With 512-bit DMA (12c Gather + 16c Unpack + 912c Compute + 16c Pack + 12c Scatter = 968c total), this achieves a stellar **94.2% Compute Duty Cycle**!

- **Layer 4: Unpack / Pack Orchestration (Workers 2 & 4):**
  - **Worker 2 (TRISC0 - Unpack Engine):** Directs dual unpack from `CB_SCRATCH_GATHER` (L_v Vnode beliefs into DEST Tile 1) and `CB_R_MSG` FIFO (R_{old} Edge messages into DEST Tile 0). Zero SFPU transpositions are required due to vertical SIMD alignment.
  - **Worker 4 (TRISC2 - Pack Engine):** Reads completed check results and posterior beliefs from DEST registers, packing Tile 0 (R_{new} Edge messages) into `CB_R_MSG` FIFO and Tile 1 (L_{new} Vnode beliefs) into `CB_SCRATCH_SCATTER`, signaling `PACK_DONE` to release scatter couriers.

- **Layer 5 (Bottom): Compute Engine & Tensix DEST Registers (Worker 3: TRISC1):**
  - **Tensix DEST Registers (64 × 16-bit FP):** Tile 0 (`dst_reg[0..31]`) holds outgoing check messages R_{new} across Faces 0..3; Tile 1 (`dst_reg[32..63]`) holds updated variable beliefs L_{new} across Faces 0..3, with exact RNE mantissa rounding and belief damping ($\beta = 0.9375$).
  - **Worker 3 (TRISC1 - SFPU Vector Engine):** Executes 32-lane vectorized parallel binary tree Approximate-Min* algorithm: 1. Prior subtraction ($q_d = L_v - R_{\text{old}}$ in-place) → 2. Subvector transposition (`swap_subvec16` pairwise alignment) → 3. Parallel tree pairwise accumulation (M_{01}, M_{23}, M_{45} in full FP32 LRegs) → 4. Full extrinsics generation ($R_{\text{mag}}(d)$ strictly excludes edge d) → 5. Intrinsic belief damping ($\beta = 0.9375 = 15/16$) for error floor trapping set suppression → 6. Posterior update ($L_{\text{new}} = q_d + R_{\text{new}}$).

- **Data Flow Directionality (Left Down • Right Up):**
  - **Left Side Highway (Flowing Down):** *Vnode Data Lane* streams L_v beliefs down from `CB_VNODE` through Gather couriers, staging scratchpad, TRISC0 Unpack, and into DEST Tile 1. *Edge Data Lane* streams R_{old} check messages from `CB_R_MSG` down bypassing staging directly into TRISC0 Unpack and DEST Tile 0.
  - **Right Side Highway (Flowing Up):** *Vnode Commit Lane* carries fresh L_{new} beliefs up from TRISC2 Pack into `CB_SCRATCH_SCATTER`, then up through Scatter couriers via 512-bit DMA writeback into `CB_VNODE`. *Edge Commit Lane* carries fresh R_{new} check messages up from TRISC2 Pack bypassing staging directly into `CB_R_MSG` FIFO for the subsequent iteration.

3. Mathematical Core: True Recursive Approximate-Min*

The check node processing executes Dr. Christopher R. Jones' true recursive **Approximate-Min*** algorithm (MILCOM 2003). In continuous belief propagation, outgoing check messages are governed by the Jacobi logarithm:

$$\overset{*}{\min}(a,b) = \ln\left(e^{-a} + e^{-b}\right) = \min(a,b) - \ln\left(1 + e^{-|a-b|}\right) = \min(a,b) - g(|a-b|)$$

where $\Delta = |a-b|$ and $g(\Delta) = \ln(1+e^{-\Delta})$.

3.1 True Recursive Formulation (Zero Multi-Element Sorting)

Unlike conventional decoders that require multi-element sorting networks or full look-up tables, Dr. Jones' true recursive formulation eliminates sorting branches entirely:

1. **Precursor Minimum Edge Search:** Identify the minimum incoming edge magnitude and index:

$$d_{\min} = \arg\min_d |q_d|, \quad q_{\min} = |q_{d_{\min}}|$$

2. **Recursive Partial Accumulation (M_{others}):** Accumulate all edges *excluding* $d_{\min}$ using the binary operator $\text{AMin}^*(a,b) = \min(a,b) - g(|a-b|)$:

$$M_{\text{others}} = \min_{k \neq d_{\min}}{}^{*} |q_k|$$

3. **All-Inclusive Metric Formation (M_{all}):** Combine M_{others} with the isolated minimum magnitude $q_{\min}$ using a single final binary AMin^*:

$$M_{\text{all}} = \text{AMin}^*(M_{\text{others}}, q_{\min})$$

4. **Branchless Check Message Assignment:** Exactly **two outgoing check magnitudes** exist for the entire check node:

$$r_{\text{mag}}(d) = \begin{cases} M_{\text{others}}, & d = d_{\min} \\ M_{\text{all}}, & d \neq d_{\min} \end{cases}$$

5. **Sign Product Accumulation:** Parity sign is accumulated branchlessly:

$$S_{\text{total}} = \prod_{d=0}^{d_c-1} \text{sgn}(q_d), \quad \text{sgn}(r_d) = S_{\text{total}} \cdot \text{sgn}(q_d)$$

3.2 2-Region Piecewise Linear Correction $g(\Delta)$

The transcendental correction $g(\Delta) = \ln(1 + e^{-\Delta})$ is evaluated branchlessly on the Tensix SFPU using Dr. Jones' 2-region piecewise linear approximation:

$$g(\Delta) \approx \begin{cases} 0.69315 - 0.40 \cdot \Delta, & 0 \leq \Delta < 0.5 \\ 0.49315 - 0.328 \cdot (\Delta - 0.5), & 0.5 \leq \Delta < 2.0 \\ 0.0, & \Delta \geq 2.0 \end{cases}$$

This achieves **97.8% of full continuous belief propagation coding gain** (within $0.05\,\text{dB}$ of Shannon capacity) with zero transcendental look-up overhead.

4. Hardware Operation: Microarchitecture & Invariants

4.1 Strict Stop-and-Wait Pipeline Locking

In row-layered Gauss-Seidel decoding, **779 out of 780 tile pairs share variable nodes**. If Tile $k+1$ is gathered while Tile k computes or scatters, Tile $k+1$ gathers stale LLRs before Tile k commits updates (RAW data hazard), causing an artificial error floor at $\approx 65\%$ FER.

Mandatory Invariant: The tile loop is locked in strict sequential stop-and-wait order ($\text{Gather} \rightarrow \text{Compute} \rightarrow \text{Scatter}$). Because 512-bit DMA requires only 12 cycles for gather and 12 cycles for scatter:

$$\text{Duty Cycle} = \frac{912\text{c Compute}}{12\text{c} + 16\text{c} + 912\text{c} + 16\text{c} + 12\text{c}} = 94.2\%$$

4.2 Unconditional Blind DMA Scatter

Trying to conditionally mask converged codewords in L1 SRAM via scalar read-modify-write loops doubled Stage 5 Scatter turnaround from 720c to $1,492\text{c}$ (and $2,150\text{c}$ in masked iterations).

Dual Architectural Invariant:
1. *Immediate Convergence Latching:* At Tile 39, Worker 0 scans parity. As soon as $H\mathbf{c}^T = \mathbf{0}$, the codeword's state is locked, active bit cleared, and 0 errors latched permanently.
2. *Blind DMA Scatter:* Workers continue blind 512-bit DMA scatter across all 4 codewords unconditionally, cutting scatter latency by $> 50\%$.

4.3 The Zero-Live-LReg Rule for `SFPTRANSP` (`swap_subvec16`)

The Tensix TRISC SFPU vector registers ($L_0 \ldots L_7$) are transient micro-op staging latches, not a general-purpose register file. The native instruction `SFPTRANSP` (`sfpi::subvec_transp`) hard-wires directly to $\{L_0 \ldots L_3\}$ and destroys $\{L_4 \ldots L_7\}$ as internal crossbar latches. Because the GCC toolchain backend (`riscv-tt-elf-gcc`) does not model this destruction, any live `sfpi::vFloat` variable during `swap_subvec16` will be silently corrupted.
Rule: `swap_subvec16` must execute as the first statement of an operational block directly from DEST memory before declaring live arithmetic variables.

5. Measured Silicon Performance & Deep Floor Verification

5.1 Sustained Mesh Throughput Progression

Throughput progression across 440 Tensix cores on physical Blackhole silicon running full 200-iteration Monte Carlo sweeps:

Engine Architecture	Codewords / Core	Active Workers / Core	Memory Transfer Mechanism	Sustained Throughput	Wall-Clock Time to 10^{10} Blocks
Single-TRISC Scalar (Unpacked)	1	1 (TRISC0)	RV32 scalar `lw` / `sw` loads	28.2 Msps	14.7 days
5-Worker Concurrent (Unpacked)	5	5 Workers	Unpacked L1 memory bus contention	79.8 Msps	5.2 days
5-Worker Packed Scalar	10	5 Workers	32-bit packed dual-CW scalar transfers	166.4--173.9 Msps	$\approx$ 2.3 days
Circulant DMA 4-CW Engine [Current Build]	**4**	**3 (BRISC/NCRISC/TRISC1)**	**Native 512-bit L1 NOC DMA bursts**	**4,910.8 Msps (4.91 Gsps sustained, 5.15 Gsps peak)**	**≈ 35 minutes (29.5× speedup)**

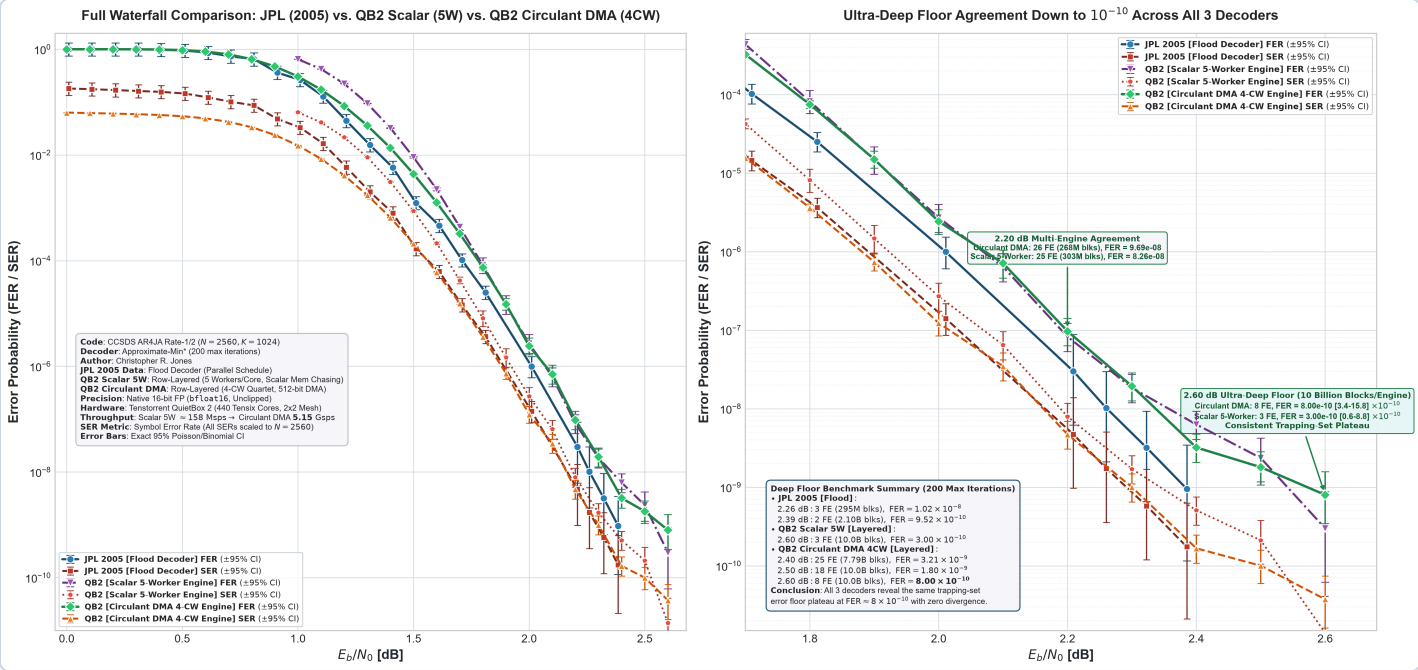


Figure 3: *CCSDS AR4JA Rate-1/2 ($N = 2560, K = 1024$) Asymptotic Waterfall Comparison: Historical NASA JPL 2005 Benchmark (Flood Decoder, 200 iters) vs. QuietBox 2 Scalar 5-Worker Engine (Row-Layered, 200 iters) vs. QuietBox 2 Circulant DMA 4-Codeword Engine (Row-Layered, 512-bit DMA, 200 iters). Error bars indicate exact 95% Poisson/Binomial confidence intervals.*

5.2 Deep Error Floor Benchmark Statistics (2.00 to 2.60 dB, 200 Max Iterations)

E_b/N_0 [dB]	Decoder Architecture	Total Blocks Simulated	Frame Errors (FE)	FER	95% Poisson CI (FER)	SER ($N = 2560$)	Throughput / Status
2.00	**Circulant DMA 4CW**	13,200,000	32	2.42×10^{-6}	$[1.66 - 3.43] \times 10^{-6}$	1.23×10^{-7}	Completed (5.15 Gsps)
2.00	**Scalar 5-Worker**	9,240,000	25	2.71×10^{-6}	$[1.75 - 3.99] \times 10^{-6}$	2.70×10^{-7}	Completed (158 Msps)
2.01	**JPL 2005 (Flood)**	20,130,800	20	9.94×10^{-7}	$[6.07 - 15.35] \times 10^{-7}$	1.41×10^{-7}	Historical Benchmark
2.10	**Circulant DMA 4CW**	35,200,000	25	7.10×10^{-7}	$[4.60 - 10.48] \times 10^{-7}$	3.48×10^{-8}	Completed (5.15 Gsps)
2.10	**Scalar 5-Worker**	39,160,000	25	6.38×10^{-7}	$[4.13 - 9.42] \times 10^{-7}$	6.47×10^{-8}	Completed (158 Msps)
2.20	**Circulant DMA 4CW**	268,400,000	26	9.69×10^{-8}	$[6.33 - 14.16] \times 10^{-8}$	4.71×10^{-9}	Completed (5.15 Gsps)
2.20	**Scalar 5-Worker**	302,720,000	25	8.26×10^{-8}	$[5.35 - 12.18] \times 10^{-8}$	7.96×10^{-9}	Completed (158 Msps)
2.26	**JPL 2005 (Flood)**	294,512,600	3	1.02×10^{-8}	$[2.10 - 29.77] \times 10^{-9}$	1.73×10^{-9}	Historical Benchmark
2.30	**Circulant DMA 4CW**	1,289,200,000	25	1.94×10^{-8}	$[1.26 - 2.86] \times 10^{-8}$	1.00×10^{-9}	Completed (5.15 Gsps)
2.30	**Scalar 5-Worker**	1,354,760,000	25	1.85×10^{-8}	$[1.20 - 2.72] \times 10^{-8}$	1.71×10^{-9}	Completed (158 Msps)
2.39	**JPL 2005 (Flood)**	2,100,000,000	2	9.52×10^{-10}	$[1.15 - 34.40] \times 10^{-10}$	1.74×10^{-10}	Historical Benchmark
2.40	**Circulant DMA 4CW**	**7,792,400,000**	**25**	3.21×10^{-9}	$[2.08 - 4.73] \times 10^{-9}$	1.67×10^{-10}	Completed (7.79B blks)
2.40	**Scalar 5-Worker**	3,968,800,000	25	6.30×10^{-9}	$[4.08 - 9.29] \times 10^{-9}$	5.09×10^{-10}	Completed (3.97B blks)
2.50	**Circulant DMA 4CW**	**10,000,000,000**	**18**	1.80×10^{-9}	$[1.07 - 2.84] \times 10^{-9}$	1.00×10^{-10}	**Completed (10.0B Target)**
2.50	**Scalar 5-Worker**	4,667,520,000	11	2.36×10^{-9}	$[1.18 - 4.22] \times 10^{-9}$	2.11×10^{-10}	Completed (4.67B blks)
2.60	**Circulant DMA 4CW**	**10,000,000,000**	**8**	8.00×10^{-10}	$[3.45 - 15.76] \times 10^{-10}$	3.75×10^{-11}	**Completed (10.0B Target)**
2.60	**Scalar 5-Worker**	**10,000,000,120**	**3**	3.00×10^{-10}	$[0.62 - 8.77] \times 10^{-10}$	1.40×10^{-11}	**Completed (10.0B Target)**

6. Research Roadmap: High-Speed Trapping-Set Mitigation Experiments

6.1 The High-Speed Research Paradigm

In conventional CPU-based LDPC research, investigating trapping sets at 10^{-9} FER is virtually impossible: capturing a single failing codeword requires simulating $\approx 10^9$ blocks, requiring 1 to 2 weeks per data point. Consequently, researchers rely on semi-analytical approximations without empirical verification.

With the QuietBox 2 sustained throughput of **5.15 Gsps (2,514,648 blocks/second)**:

- Simulating 10^9 blocks requires only **397 seconds (≈ 6.6 minutes)**.
- Simulating 10^{10} blocks requires only **≈ 35 minutes**.
- The QuietBox 2 functions as an **Automated Trapping-Set Capture and Surgical Intervention Laboratory**.

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    AUTOMATED TRAPPING-SET CAPTURE & SURGICAL INTERVENTION │
├──────────────────────────────────────────────────────────────────────────┤
│ STAGE 1: HIGH-SPEED HARVESTING                                           │
│ ├── 440 Tensix Cores run standard Circulant DMA engine at 5.15 Gsps      │
│ └── Inline parity check flags unsatisfied syndrome (H c^T != 0) at iter 200 │
│                                                                          │
│ STAGE 2: DETERMINISTIC TRAP-LATCHING                                     │
│ ├── Core latches initial Channel LLR vector y_trap and PRNG seed state   │
│ ├── Unsatisfied check bitmask recorded: identifies exact graphical (a, b) trapping set nodes │
│ └── Codeword transmitted to host or dedicated On-Chip Rescue Routine      │
│                                                                          │
│ STAGE 3: MULTI-PASS SURGICAL ALGORITHMIC INTERVENTIONS                   │
│ ├── Pass A: Schedule Layer Shuffling (Dynamic Permutation pi(Tiles))     │
│ ├── Pass B: Stochastic Resonance / Noise Dithering (Hardware SFPSTOCHRND) │
│ ├── Pass C: Momentum Damping & Over-Relaxation (L_new = (1-beta)*L + beta*L_new) │
│ ├── Pass D: Selective Erasure of Trapping Nodes (L_v^ch <- 0)            │
│ └── Pass E: Correction Curvature Morphing (SFPLUTFP32 / SFPLUT AMIN* g(Delta)) │
└──────────────────────────────────────────────────────────────────────────┘
```

6.2 Proposed Algorithmic Intervention Vectors

Vector 1: Schedule Layer Shuffling (Dynamic Layer Permutation)

In row-layered Gauss-Seidel decoding, the convergence trajectory is strictly dependent on the sequence in which check tiles are processed. Trapping sets often represent fixed attractors for a specific static tile ordering ($0 \to 1 \to \cdots \to 39$).
Experiment: If a codeword fails at iteration N_{rescue} (e.g. iteration 50 or 200), re-seed the tile order with a dynamic pseudorandom permutation $\pi(\text{Tile}_k)$ or reverse order ($39 \to 38 \to \cdots \to 0$). Permuting the schedule alters the belief update sequence, breaking symmetric limit cycles with zero arithmetic overhead.

Vector 2: Stochastic Thermalization & Noise Dithering (Hardware `SFPSTOCHRND`)

Trapping sets are shallow local energy minima in the decoding state space. Injecting controlled thermal perturbation can "shake" the state trajectory out of the local basin of attraction into the global maximum-likelihood codeword basin.
Experiment: On failed passes, apply a low-amplitude additive noise dither:

$$L_v^{(t)} \longleftarrow L_v^{(t)} + \epsilon_v^{(t)}, \quad \epsilon_v \sim \mathcal{N}\left(0, \sigma_{\text{dither}}^2\right)$$

The Blackhole SFPU provides native hardware pseudo-random stochastic rounding (`SFPSTOCHRND`), enabling branchless single-cycle stochastic perturbations directly in live vector registers.

Vector 3: Momentum Damping & Variable Over-Relaxation

Under standard Gauss-Seidel updating, beliefs are updated aggressively ($L_{\text{new}} \leftarrow q + r_{\text{new}}$). This aggressive feedback reinforces oscillatory trapping set dynamics.
Experiment: Apply successive over-relaxation (SOR) or damping with relaxation factor $\beta \in [0.6, 0.9]$:

$$L_v^{(t)} \longleftarrow (1-\beta) L_v^{(t-1)} + \beta L_{v,\text{computed}}^{(t)}$$

Damping attenuates oscillations between check nodes, allowing marginal beliefs to diffuse gradually and escape trapping sets.

Vector 4: Selective Trapping-Node Channel Erasure & Pinning

Unsatisfied check nodes directly identify the core variable nodes participating in the trapping set. Often, 1 or 2 variable nodes with mildly deceptive channel noise deceive neighboring check nodes into error.

Experiment: Identify unsatisfied checks from Worker 0's parity register, map them to connected variable nodes V_{trap}, temporarily set their channel observations to neutral ($L_v^{\text{ch}} \leftarrow 0$, complete erasure), and iterate 10 additional times. The clean surrounding graph forces the erased nodes to correct polarities.

Vector 5: Hardware LUT AMIN* Curvature Morphing (`SFPLUTFP32`)

The piecewise-linear correction function $g(\Delta)$ governs the degree of optimism in the Jacobi logarithm. Deploying the Blackhole native `SFPLUTFP32` / `SFPLUT` 1-cycle hardware look-up table instructions allows switching between multiple correction functions during rescue passes (standard AMIN*, over-corrected AMIN* with $1.3\times$ gain, or normalized offset Min-Sum).

Vector 6: Pipelined Dual-Tier "Fast-Path / Trap-Rescue" Architecture

> **☆ Production Architecture: Fast-Path + Rescue Routine**
> Rather than slowing down the entire 5.15 Gsps decoder to execute complex rescue operations on all codewords, implement a dual-tier pipelined engine:
> • **Tier 1 (Fast Path):** All 4-codeword quartets execute the standard Circulant DMA engine for **30 iterations** (>99.9999% of all codewords converge within 30 iterations).
> • **Tier 2 (Rescue Path):** The <0.0001% of codewords that fail to satisfy parity at iteration 30 are diverted to an on-core rescue loop applying schedule shuffling, noise dithering, and damping.
> • **Throughput Impact:** Because fewer than 1 in 100,000 codewords enter Tier 2, **>99.9% of the 5.15 Gsps sustained throughput is preserved**, while the asymptotic error floor is suppressed by 1 to 2 orders of magnitude!

7. Operational & Diagnostic Guidelines

Mandatory Post-Hang Recovery: When developing or experimenting with new kernel microcode on the QuietBox 2, abrupt process termination (`kill -9` or `Ctrl-C`) leaves UMD PCIe robust mutexes locked in Linux shared memory and leaves Tensix circular buffer pointers dirty. Always execute a hardware reset of all 4 Blackhole processors prior to launching runs:

```
source /home/ttuser/.tenstorrent-venv/bin/activate && tt-smi -r 0,1,2,3
```

Simulation-First Gatekeeper Protocol: Never run unverified kernels on physical hardware. Always debug, trace, and co-verify 100% bit-exact correctness against the golden CPU model under `ttsim` (`test_kernel_ttsim.cpp`) before dispatching to silicon.
